# Algorithmic Explanation and Phenomenal Consciousness: A Conditional Impossibility Result

Luca Martijn Goddijn
Independent Researcher
Amsterdam, Netherlands

## Abstract

This paper develops a formal analysis of a conditional limitation on explaining consciousness. If no algorithmic theory can achieve complete strong capture of phenomenal consciousness, then a complete algorithmic explanation is impossible. We refine this claim by distinguishing epistemic, conceptual, and computational limits, introducing a formal notion of representability, and incorporating empirical constraints from neuroscience. The weaker operational premise on which the theorem depends is supported by three independently motivated constraints — representability, algorithmic realizability, and explanatory closure — whose convergent failure is shown to be evidentially asymmetric: more naturally explained by a principled limitation than by coincidental gaps. We further analyze the algorithmic status of human reasoning itself, showing that it remains formally underdetermined. This underdetermination places structural limits on the scope of explanation and strengthens the conditional nature of the main theorem.

## Keywords

phenomenal consciousness, hard problem, algorithmic explanation, explanatory gap, computational theory of mind, artificial consciousness, consciousness attribution, meta-theory

## 1 Introduction

The hard problem of consciousness concerns the relation between physical processes and subjective experience, and the associated explanatory gap between them [38]. While empirical research has made substantial progress in identifying neural correlates, it remains unclear whether such progress contributes to explaining why experience exists at all.

This paper develops a conditional impossibility result concerning complete strong explanation of phenomenal consciousness. Rather than claiming that consciousness is straightforwardly non-algorithmic, the argument is that no algorithmic theory has been shown capable of both explaining why phenomenal consciousness exists and covering the full set of truths about it. While the theorem itself requires only this weaker premise, the paper also examines the evidential relationship between the operational premise and the stronger principled thesis that phenomenal consciousness is not algorithmically capturable, arguing that the pattern of evidence supporting the weaker claim is not neutral between the two possibilities. The result is situated within a layered framework distinguishing epistemic, conceptual, computational, instantiational, and attributional constraints. A further layer is added through analysis of the nature of human reasoning itself, which underlies all admissible explanations. The formal vocabulary developed here is meta-theoretic: it does not compete with object-level theories of consciousness such as Integrated Information Theory or Global Workspace Theory, but provides a framework for evaluating what any such theory achieves in terms of explanatory completeness. The paper also argues that the distinction between predictive success and phenomenal instantiation is not merely a formal one: contemporary artificial systems already elicit widespread attributions of consciousness on the basis of behavioral fluency alone, and this creates a practical risk of conflating attribution with explanation.

## 2 First-Person vs Third-Person Access

Any theory of consciousness faces an initial asymmetry of access. On the one hand, conscious experience is available from the first-person point of view through introspection; on the other hand, mechanisms and correlates are available from the third-person point of view through observation and experiment.

The total epistemic situation may be represented schematically as:

$$\mathcal{E}_{\text{access}} = E_{int} \cup E_{ext}$$

where:

- $E_{int}$ denotes first-person or introspective access,
- $E_{ext}$ denotes third-person or observational access.

The crucial limitation is that neither mode of access, taken alone, provides unrestricted access to the full body of truths about consciousness:

$$E_{int} \not\supseteq \text{Truth}(C), \qquad E_{ext} \not\supseteq \text{Truth}(C)$$

This asymmetry motivates the later distinction between empirical progress and complete explanation [11, 44, 60, 65].

## 3 Epistemic vs Computational Limits

It is important to distinguish two kinds of limitation that are often conflated in discussions of consciousness.

- **Epistemic limitation:** a limit on what can be known, accessed, or observed.
- **Computational limitation:** a limit on what can be formally derived or algorithmically captured.

The distinction matters because an epistemic gap does not automatically imply a computational one, and vice versa. A phenomenon may be difficult to access but still formally capturable, or formally intractable despite being directly experienced.

The framework developed in this paper concerns both kinds of limitation, but gives special attention to cases in which epistemic, representational, and computational constraints converge [15, 21, 64].

## 4 Conceptual Representability

Let $\mathcal{R}$ denote the set of structures that admit formal representation.

$$C \in \mathcal{R} \iff C \text{ admits a formal representation}$$

We assume:

$$\mathcal{A} \subseteq \mathcal{R}$$

That is, anything algorithmically capturable must first be representable. Hence:

$$C \notin \mathcal{R} \Rightarrow C \notin \mathcal{A}$$

*Definition of algorithmic theories.* In this paper, $\mathcal{A}$ denotes the class of theories whose explanatory relations are *Turing-computable.* That is, a theory $T \in \mathcal{A}$ if the mappings, derivations, or procedures by which $T$ relates explanans to explanandum can be implemented by a Turing machine.

This notion is intentionally weaker than full formal axiomatizability. The requirement is not that a theory be expressible within a fixed formal system, but that its explanatory content be effectively computable. This aligns the notion of algorithmic explanation with the standard interpretation of effective procedure in computability theory.

This notion becomes critical later when considering whether the full truth-set $\mathrm{Truth}(C)$ can be brought within the scope of a formal explanatory theory. If some aspect of consciousness is not formally representable, then no algorithmic theory can completely explain it [12, 30, 40].

## 5 Composition and Instantiation Limits

A central issue in theories of consciousness is whether any single explanatory framework can capture all relevant aspects of phenomenal consciousness. Let $\mathcal{F}$ denote the class of candidate explanatory frameworks, where a framework may be compositional, reductive, functional, biological, computational, or otherwise.

For any framework $F \in \mathcal{F}$, say that $F$ *fails on* a domain $P \subseteq \mathrm{Truth}(C)$ if $F$ does not adequately represent, account for, or explain the truths in $P$.

The general limitation may then be stated schematically as:

$$\forall F \in \mathcal{F},\ \exists P \subseteq \mathrm{Truth}(C) \text{ such that } F \text{ fails on } P$$

This expresses the claim that no single explanatory framework has yet been shown to exhaust the full truth-set of consciousness.

Two recurrent forms of incompleteness are especially relevant.

- **Compositional limitation:** a framework may explain relations among parts, contents, correlates, or mechanisms while failing to explain why those organized structures are accompanied by experience.
- **Instantiational limitation:** a framework may specify conditions under which consciousness is said to be present, while failing to explain why those conditions yield phenomenal consciousness rather than merely functional or structural organization.

In this sense, compositional frameworks tend to fall short at the level of phenomenal explanatory closure, while instantiational frameworks often fall short at the level of complete explanatory articulation. This is one reason complete strong capture requires more than success in any single explanatory dimension [25, 41, 42, 58].

## 6 Neural Structure of Conscious Content

Empirical work suggests that consciousness is neither simple nor localized, but structured and distributed. Phenomenal consciousness may therefore be treated, at least schematically, as internally differentiated:

$$C = (c_1, c_2, ..., c_n)$$

where the $c_i$ denote distinguishable components, dimensions, or aspects of conscious content.

Let

$$\phi : C \to \mathcal{N}$$

map conscious contents to their neural correlates or realizations. At minimum, distinct conscious contents need not collapse onto a single neural state:

$$\phi(c_i) \neq \phi(c_j) \quad \text{for at least some } i \neq j$$

This suggests that consciousness is distributed across multiple neural structures rather than reducible to a single unitary correlate. In particular, there need not exist a single neural item $N \in \mathcal{N}$ whose inverse image under $\phi$ recovers the whole structured field of conscious content:

$$\nexists N \in \mathcal{N} \text{ such that } \phi^{-1}(N) = C$$

This is a schematic way of expressing distributed realization, not a claim that current neuroscience has established a mathematically exact map of this form. The point is that local neural correlates do not by themselves amount to a complete capture of the phenomenal field [16, 24, 65].

## 7 Separation of Empirical and Conceptual Problems

The explanatory problem of consciousness contains at least two distinct targets:

- **Empirical target:** identifying mechanisms, correlates, and functional organization.
- **Conceptual target:** explaining why and how phenomenal experience exists at all.

These targets should not be conflated [11, 12, 64]. Formally:

$$\text{Explaining } \mathcal{N} \not\Rightarrow \text{Explaining } C$$

Let $\mathrm{Func}(C)$ denote the functional or access-related description of consciousness. Functional description concerns the role a state plays in reasoning, report, control, and information availability.

At minimum:

$$\mathrm{Func}(C) \subseteq \mathrm{Truth}(C)$$

since functional truths form part of the broader body of truths about consciousness.

Whether functional description exhausts $\mathrm{Truth}(C)$ depends on whether phenomenal consciousness is reducible to access or functional organization. If phenomenal consciousness is not reducible in that way, then:

$$\mathrm{Func}(C) \neq \mathrm{Truth}(C)$$

The following section motivates precisely this possibility through the distinction between phenomenal and access consciousness.

Accordingly,

$$\mathrm{Progress}(\mathcal{N}) \not\Rightarrow \mathrm{Progress}(\mathrm{Truth}(C))$$

Progress in identifying correlates or mechanisms does not by itself settle the question of whether phenomenal consciousness has been explained.

## 8 Phenomenal vs Access Consciousness

A central clarification concerns the nature of consciousness itself. Following Block [6], consciousness is not a unitary concept but divides into at least two distinct notions:

- **Phenomenal consciousness**: subjective experience, or what it is like to be in a state
- **Access consciousness**: availability of information for reasoning, report, and control of behavior

Block argues that these are conceptually distinct. At minimum, this implies:

$$C_{phen} \neq C_{access}$$

and motivates the possibility of dissociation:

$$\exists s \text{ such that } s \in C_{phen} \wedge s \notin C_{access}$$

This distinction underlies the claim that functional or computational descriptions of cognition may fail to capture subjective experience.

### 8.1 The Overflow Debate

Block's overflow argument [7] suggests that phenomenal consciousness may exceed cognitive access. Empirical paradigms such as Sperling's partial report task indicate that subjects may have richer perceptual experience than can be reported or maintained in working memory.

This motivates a structural inequality:

$$\mathrm{Dim}(C_{phen}) > \dim(C_{access})$$

However, this claim remains debated. Yang [66] argues that evidence interpreted as supporting fragmentation may also be consistent with rich phenomenology.

### 8.2 Neural Dissociation

Neuroscientific accounts such as recurrent processing theory [36] propose distinct neural mechanisms for phenomenal and access consciousness. Early recurrent processing in sensory areas may support phenomenal experience, while later global ignition supports access.

This suggests:

$$\phi(C_{phen}) \neq \phi(C_{access})$$

indicating a possible separation at the level of neural realization.

### 8.3 Access-Based Theories

Opposing views argue that phenomenal consciousness reduces to access. Overgaard [46] and Naccache [43] contend that existing empirical evidence does not establish a clear separation. Cohen and Dennett [17] argue that consciousness is fundamentally functional and cannot be separated from cognitive processes.

Thus, the distinction between phenomenal and access consciousness remains an open question.

### 8.4 Working Definition of $C$

For the purposes of this paper, we define:

$$C := C_{phen}$$

That is, $C$ denotes phenomenal consciousness: the aspect of experience not exhausted by functional accessibility.

This restriction aligns the target of the present analysis with the explanatory gap and the hard problem of consciousness, rather than with cognitive or behavioral capacities alone.

## 9 Conditions of Explanation and Capture

The central claim of this paper concerns the limits of explaining consciousness. To state that claim precisely, it is necessary to distinguish different strengths of explanatory success.

A theory may relate to consciousness in more than one way. It may correlate with conscious states, reproduce aspects of their internal structure, or account for why there is experience at all. These are not equivalent achievements.

### 9.1 Truths About Consciousness

Let $\mathrm{Truth}(C)$ denote the totality of *non-explanatory* truths about phenomenal consciousness. At minimum, this includes:

- truths about the occurrence of phenomenal states,
- truths about the internal structure of conscious content,
- truths about relations between phenomenal states and their neural or functional correlates.

We distinguish these from a further class of potential truths:

$$\mathrm{Truth}_{\mathrm{why}}(C)$$

which, if they exist, concern why phenomenal consciousness accompanies certain physical or functional states at all.

The status of $\mathrm{Truth}_{\mathrm{why}}(C)$ is philosophically contested. Some positions hold that such truths exist and are in principle explainable; others treat consciousness as a fundamental feature of reality requiring no further explanation.

Accordingly, the requirement of explanatory closure presupposes that $\mathrm{Truth}_{\mathrm{why}}(C)$ is non-empty. If no such truths exist, then the requirement of complete explanation becomes vacuously unsatisfiable. In that case, the limitation result holds trivially, not because of a failure of algorithmic theories, but because the explanatory demand has no corresponding target.

## 9.2 Levels of Capture

Three levels of capture should be distinguished.

*Weak capture.* A theory $T$ weakly captures consciousness if it reliably tracks, predicts, or correlates with conscious states:

$$\mathrm{Capture}_{\mathrm{w}}(T, C)$$

This includes the identification of neural correlates, behavioral signatures, or functional markers of consciousness.

*Structural capture.* A theory $T$ structurally captures consciousness if it reproduces or preserves significant aspects of the internal organization of conscious content:

$$\mathrm{Capture}_{\mathrm{s}}(T, C)$$

This is stronger than mere prediction. A structurally capturing theory does not merely detect whether consciousness is present, but models relevant relations among its contents, dimensions, or transitions.

*Explanatory closure.* A theory $T$ achieves explanatory closure with respect to consciousness if it explains why phenomenal consciousness exists at all, rather than merely correlating with it or reproducing its structure:

$$\mathrm{ExplClosure}(T, C)$$

This is the explanatory demand characteristic of the hard problem.

*Strong capture.* A theory $T$ strongly captures consciousness if it achieves explanatory closure and its explanatory scope includes the relevant truths about consciousness:

$$\mathrm{Capture}_{\mathrm{str}}(T, C) \iff \mathrm{ExplClosure}(T, C) \land \mathrm{Truth}(C) \subseteq \mathrm{Scope}(T)$$

where $\mathrm{Truth}(C)$ denotes non-explanatory truths, and $\mathrm{ExplClosure}(T, C)$ accounts for the explanatory component corresponding to $\mathrm{Truth}_{\mathrm{why}}(C)$ when such truths exist.

Strong capture is therefore stronger than both weak and structural capture. It is the notion relevant to the present paper's analysis of complete explanation.

## 9.3 Relations Between the Levels

These notions stand in a hierarchical relation:

$$\mathrm{Capture}_{\mathrm{str}}(T, C) \Rightarrow \mathrm{Capture}_{\mathrm{s}}(T, C) \Rightarrow \mathrm{Capture}_{\mathrm{w}}(T, C)$$

but not conversely.

These implications follow from the definitions. If a theory strongly captures consciousness, then it must already succeed in preserving the relevant structure of conscious content and in reliably tracking the states in question. Likewise, if a theory structurally captures consciousness, then it must already succeed at weak capture, since one cannot preserve relations among conscious contents without first identifying or tracking those contents.

The converses fail. Reliable correlation does not by itself explain why experience exists, and structural preservation does not by itself yield phenomenal explanatory closure.

Remark 1. *The implication* $\mathrm{Capture}_{\mathrm{str}}(T, C) \Rightarrow \mathrm{Capture}_{\mathrm{s}}(T, C)$ *holds because strong capture requires truth-completeness:* $\mathrm{Truth}(C) \subseteq \mathrm{Scope}(T)$. *Since* $\mathrm{Truth}(C)$ *includes truths about the internal structure of conscious content — the spatial organization of the visual field, the qualitative differences among sensory modalities, the relations among phenomenal states — any theory whose scope includes these truths must preserve significant aspects of that structure. A theory that explains why consciousness exists without modeling its structure — for example, by positing consciousness as a fundamental property of reality — may achieve explanatory closure but not truth-completeness, and therefore does not achieve strong capture. The hierarchy is grounded in the definition of strong capture rather than in an independent assumption about the relationship between explanation and structural description.*

## 9.4 Explanation and the Hard Problem

The distinction above allows us to sharpen the separation between empirical and conceptual progress. Empirical research may achieve weak and, in some cases, structural capture of consciousness. For example, a theory may identify neural correlates or characterize distributed content representations. But this does not by itself yield strong capture.

Formally:

$$\mathrm{Capture}_{\mathrm{w}}(T, C) \nRightarrow \mathrm{Capture}_{\mathrm{str}}(T, C)$$

and

$$\mathrm{Capture}_{\mathrm{s}}(T, C) \nRightarrow \mathrm{Capture}_{\mathrm{str}}(T, C)$$

This restates the earlier result:

$$\text{Explaining } \mathcal{N} \nRightarrow \text{Explaining } C$$

The mere existence of a neural or functional account does not settle the question of why phenomenal consciousness is present [11, 12, 44].

## 9.5 Why Explanatory Closure is Structurally Resistant

A central difficulty concerns the nature of phenomenal explanation itself. Physical and functional descriptions operate in a third-person, structural vocabulary: they specify relations, mechanisms, and causal organization. Phenomenal consciousness, by contrast, is essentially first-personal and perspectival: it concerns what it is like to be in a given state.

This suggests a structural mismatch. Let:

$$\mathcal{D}_{\mathrm{struct}} \quad \text{and} \quad \mathcal{D}_{\mathrm{phen}}$$

denote the domains of structural and phenomenal description, respectively. Then explanatory closure requires a mapping:

$$f : \mathcal{D}_{\mathrm{struct}} \to \mathcal{D}_{\mathrm{phen}}$$

that not only correlates structures with experiences, but explains why the former are accompanied by the latter.

However, phenomenal contents appear to involve irreducibly perspectival or indexical elements. These elements are not straightforwardly eliminable in favor of third-person structural descriptions without loss of what is to be explained [12, 38, 44]. Algorithmic theories, by contrast, operate over formally representable and intersubjectively accessible structures.

Thus, explanatory closure appears to require bridging a domain that is structurally characterizable with one that is essentially perspectival. No generally accepted method is known for eliminating this gap without either reducing phenomenal content to structure or leaving the explanatory demand unmet.

Even if this gap is ultimately cognitive rather than metaphysical, the requirement of explanatory closure remains unsatisfied at the level of available theories. This suffices, within the present framework, to block complete strong capture.

## 9.6 Worked Example: Global Workspace Theory

To illustrate the capture hierarchy, consider Global Workspace Theory (GWT), one of the most empirically supported theories of consciousness [2, 23].

GWT identifies a neural mechanism — global broadcasting of information across a network of specialized processors — and proposes that this mechanism is the basis of conscious access. In terms of the present framework:

- **Weak capture:** GWT reliably tracks which stimuli reach conscious access and which do not. It predicts, for example, that masked stimuli failing to trigger widespread cortical ignition will not be consciously perceived. This constitutes $\mathrm{Capture}_{\mathrm{w}}(T_{GWT}, C)$.
- **Structural capture:** GWT preserves aspects of the internal organization of conscious content, including the transition from local processing to global availability, the limited capacity of the workspace, and the serial character of conscious access. This constitutes $\mathrm{Capture}_{\mathrm{s}}(T_{GWT}, C)$.
- **Explanatory closure:** GWT does not explain why global broadcasting is accompanied by phenomenal experience rather than occurring without any subjective character. A system could, in principle, broadcast information globally while remaining experientially dark. This is a standard formulation of the explanatory gap applied to GWT.

Thus:

$$\mathrm{Capture}_{\mathrm{s}}(T_{GWT}, C) \;\; \text{but} \;\; \neg\,\mathrm{ExplClosure}(T_{GWT}, C)$$

and therefore:

$$\neg\,\mathrm{Capture}_{\mathrm{str}}(T_{GWT}, C)$$

The same pattern applies, with appropriate modifications, to other leading theories. Integrated Information Theory specifies a quantity ($\Phi$) associated with consciousness and characterizes its structural conditions, but does not explain why systems with high $\Phi$ feel like anything [1, 62]. Recurrent Processing Theory identifies a neural mechanism (local recurrent activity) as the basis of phenomenal consciousness, but the question of why recurrent processing gives rise to experience rather than merely to enhanced information processing remains open [36].

In each case, the theory achieves weak and structural capture while falling short of explanatory closure. This pattern motivates the distinction between levels of capture and illustrates why structural success does not by itself constitute complete explanation.

A defender of GWT may respond that global broadcasting does not merely correlate with consciousness but constitutes it. On such a view, no further explanation is required: identifying the relevant physical process suffices. Within the present framework, however, this amounts to structural capture unless it explains why this structure is accompanied by phenomenal experience rather than occurring without it. The demand for explanatory closure is not for an additional mechanism, but for an account of this constitutive relation itself.

## 9.7 Working Notion of Complete Explanation

A theory $T$ completely explains consciousness only if it achieves complete strong capture of $C$.

Equivalently, complete explanation requires both:

- phenomenal explanatory closure, and
- full coverage of the truths in $\mathrm{Truth}(C)$.

Formally, complete explanation requires:

$$\mathrm{Capture}_{\mathrm{str}}(T, C)$$

under the definition given above.

This makes explicit that the present paper does not deny the possibility of weak or structural theories of consciousness. The target is the stronger claim that phenomenal consciousness can be completely explained in the sense relevant to the hard problem.

## 9.8 Why This Distinction Matters

Without distinguishing levels of capture, the debate over explanation becomes unstable. Weak empirical success can be mistaken for full explanation, while critiques aimed at strong explanation may be incorrectly taken to deny the value of empirical work.

The framework adopted here avoids both errors:

- it preserves the legitimacy of empirical progress,
- it allows structural modeling of conscious content,
- and it isolates the stronger explanatory demand that remains philosophically contested.

Accordingly, the theorem later in this paper should be read as a claim about the impossibility of *complete strong algorithmic capture*, conditional on the stated assumptions. This distinction will also matter later when considering artificial systems, since behavioral success may invite attributions of consciousness even where strong capture has not been established.

# 10 On the Algorithmic Status of Human Reasoning

Let $M$ denote the human cognitive system responsible for generating explanations. A central question underlying the present framework is whether $M$ itself is algorithmic, i.e.:

$$\text{Is } M \in \mathcal{A}\,?$$

This question has been extensively studied across philosophy, logic, artificial intelligence, and cognitive science. However, despite decades of work, no conclusive answer has been established.

## 10.1 The Lucas–Penrose Argument

One of the most prominent arguments against the algorithmic nature of human reasoning is based on Gödel's incompleteness theorem [29]. Lucas [39] and Penrose [48, 49] argue that for any formal system $F$ intended to capture human mathematical reasoning, one can construct a Gödel sentence $G(F)$ such that:

$$G(F) \text{ is true but not provable within } F$$

They claim that human reasoners can recognize the truth of $G(F)$, and therefore:

$$M \notin \mathcal{A}$$

However, this argument has been widely criticized. A central objection is that it assumes that human reasoning is consistent and capable of reliably identifying the truth of Gödel sentences. If human reasoning is not perfectly consistent, or if our belief in the truth of $G(F)$ is fallible, the argument fails [33, 35]. Benacerraf further argues that the most that follows is a disjunction concerning formal systems and human reasoning capacity [3].

More formally, the Lucas–Penrose argument does not establish:

$$\text{Provable}(M \notin \mathcal{A})$$

but at most suggests a conditional limitation under idealized assumptions about human reasoning.

## 10.2 The Computational Theory of Mind

The opposing position is given by the Computational Theory of Mind (CTM), according to which cognitive processes are computational processes [54]. Classical formulations are associated with Putnam [51], Fodor [27], and Pylyshyn [53].

On this view:

$$M \in \mathcal{A}$$

CTM is supported by the success of computational models in explaining aspects of cognition, as well as by the Church–Turing thesis [21, 63], which informally identifies effective calculability with Turing-computability.

However, the Church–Turing thesis is not a formally provable statement [15]. Consequently, CTM remains a theoretical and empirical hypothesis rather than a proven result.

Thus, CTM does not establish:

$$\text{Provable}(M \in \mathcal{A})$$

but rather provides a framework within which cognitive processes can be modeled.

## 10.3 Semantic Objections: Searle's Argument

Searle's Chinese Room argument [57] challenges a different aspect of computationalism. Rather than denying that cognition might be computational, Searle argues that computation alone is insufficient for genuine understanding.

Formally, Searle distinguishes between:

$$\text{Syntax} \quad \text{and} \quad \text{Semantics}$$

and argues that:

$$\text{Syntax} \not\Rightarrow \text{Semantics}$$

Thus, even if:

$$M \in \mathcal{A}$$

it does not follow that computation alone suffices to explain mental content or consciousness. This does not refute computationalism, but limits its explanatory scope [18, 58].

## 10.4 Hypercomputation and Physical Possibility

Another line of argument considers whether the brain might implement forms of computation beyond the Turing limit. Siegelmann [61] showed that certain analog neural networks with real-valued weights can compute functions not computable by Turing machines.

This suggests the possibility:

$$M \notin \mathcal{A}$$

However, such models rely on idealized assumptions, such as infinite precision in real-valued parameters. Critics argue that these assumptions are not physically realizable, and therefore do not establish that biological systems are genuinely non-computable [20, 22].

Additional proposals, such as modal arguments for hypercomputing minds [8], remain speculative and lack empirical confirmation.

Thus, hypercomputation remains a theoretical possibility rather than an empirically established feature of human cognition.

## 10.5 Underdetermination of Cognitive Computability

Taken together, the above arguments lead to a central conclusion:

Proposition 2 (Underdetermination of Cognitive Computability).

$$\neg\text{Provable}(M \in \mathcal{A}) \ \wedge \ \neg\text{Provable}(M \notin \mathcal{A})$$

That is, the algorithmic status of human cognition is currently underdetermined.

## 10.6 Implications for Explanation

Let $\mathcal{E}_M$ denote the set of explanations produced by $M$. One might assume that if human cognition is non-algorithmic, then explanations themselves need not be algorithmic. However, this does not follow.

Formally:

$$M \notin \mathcal{A} \ \not\Rightarrow \ \mathcal{E}_M \not\subseteq \mathcal{A}$$

This reflects the possibility that even if human cognition involves non-algorithmic processes, the explanations it produces may still be constrained to algorithmic forms.

Conversely, if $M \in \mathcal{A}$, then:

$$\mathcal{E}_M \subseteq \mathcal{A}$$

follows naturally.

Thus, the assumption:

$$A1 : \mathcal{E} \subseteq \mathcal{A}$$

remains a substantive but defensible constraint. It reflects the idea that explanations, as communicable and formalizable objects, are subject to algorithmic structure, regardless of the underlying nature of cognition.

### 10.7 Summary

The analysis above shows that the algorithmic status of human reasoning remains unresolved. This underdetermination has two important consequences:

- It prevents any unconditional claim that explanation must be algorithmic.
- It reinforces the conditional nature of the main theorem of this paper.

Accordingly, the limitations derived in this work should be understood as contingent on assumptions about the nature of explanation, rather than as absolute constraints on cognition itself.

## 11 From A2 to A2': Status and Justification of the Weaker Premise

The stronger philosophical thesis considered in this paper may be stated as:

$$A2 : \mathrm{Truth}(C) \notin \mathcal{A}$$

This formulation asserts that the full body of truths about phenomenal consciousness is not algorithmically capturable. While this remains a coherent and philosophically significant position, it is stronger than what is required for the central limitation result developed here.

The central burden of the present paper is therefore not to derive the main theorem from first principles, but to motivate the plausibility of a weaker premise sufficient for the theorem.

For the theorem itself, a weaker and more targeted operational premise is adopted:

Let $\mathrm{Scope}(T)$ denote the set of propositions or truths expressible and addressed by theory $T$.

$$A2' : \neg\exists T \in \mathcal{A} \text{ such that } \mathrm{Capture}_{\mathrm{str}}(T, C)$$

That is, there is no algorithmic theory that both:

- achieves strong capture (explains why phenomenal consciousness exists), and
- covers the full set of truths about consciousness.

This formulation avoids the stronger metaphysical claim that $\mathrm{Truth}(C)$ is itself non-algorithmic, and instead targets the joint requirement of explanation and completeness over consciousness.

### 11.1 Decomposing Strong Capture

Under the definitions adopted in the previous section, strong capture involves two jointly necessary components:

$$\mathrm{Capture}_{\mathrm{str}}(T, C) \iff \mathrm{ExplClosure}(T, C) \wedge \mathrm{Truth}(C) \subseteq \mathrm{Scope}(T)$$

Thus, strong capture consists of:

- **Phenomenal explanatory closure** ($\mathrm{ExplClosure}(T, C)$): the theory explains why phenomenal consciousness exists,
- **Truth-completeness**: the theory covers the full set of truths in $\mathrm{Truth}(C)$.

A theory may satisfy one condition without satisfying the other. A theory may, for example, provide a candidate explanation while failing to cover the whole truth-set, or it may enumerate a large body of truths while failing to explain why experience exists at all. The present impossibility result concerns their conjunction.

### 11.2 Necessary Conditions for Complete Strong Capture

For an algorithmic theory to achieve complete strong capture, at least the following conditions must hold:

(a) **Representability:** $\mathrm{Truth}(C)$ must be formally representable:

$$\mathrm{Truth}(C) \subseteq \mathcal{R}$$

(b) **Algorithmic realizability:** the relations connecting phenomenal and physical or functional descriptions must be algorithmically derivable.

(c) **Explanatory sufficiency:** the resulting theory must explain why phenomenal consciousness exists, not merely describe its structure.

Failure of any one of these conditions is sufficient to block complete strong capture.

### 11.3 Layered Support for A2'

The plausibility of $A2'$ is supported by three independent layers of argument.

*Layer 1: Positive Constraints.* Several well-supported distinctions motivate the claim that complete strong capture is not achieved:

$$\mathrm{Truth}(C) \neq \mathrm{Func}(C)$$

$$\mathrm{Capture}_{\mathrm{s}}(T, C) \nRightarrow \mathrm{Capture}_{\mathrm{str}}(T, C)$$

$$\mathrm{Pred}(S, C) \nRightarrow \mathrm{Inst}(S, C)$$

Together, these establish that functional, structural, and behavioral success do not suffice for strong capture.

*Layer 2: Underdetermination Results.* Even if the above constraints were challenged, key links required for reduction remain unproven:

$$\neg\mathrm{Provable}(\mathrm{Struct}(S, C) \Rightarrow \mathrm{Inst}(S, C))$$

$$\neg\mathrm{Provable}(\mathrm{Truth}(C) \subseteq \mathcal{R})$$

In particular, if phenomenal truths are essentially perspectival or indexical, their totality may fail to admit a unified third-person formal representation. In that case, representability itself becomes questionable, and it cannot be established that phenomenal truths are fully formalizable or that structural reproduction yields instantiation [12, 30, 40, 44].

*Layer 3: Structural Limitation.* The inference from local computability to global computability depends on assumptions about compositional structure that may not hold for unified phenomenal fields. In particular, the unity of consciousness raises questions related to the binding problem and the combination problem, where the whole is not straightforwardly reducible to independently computable parts.

Accordingly, even if:

$$(\forall i,\ c_i \in \mathcal{A})$$

it does not immediately follow that the unified phenomenal field $C$ is algorithmically capturable as a whole.

## 11.4 Triple Constraint on Strong Algorithmic Capture

The preceding analysis can be summarized as a single structural result.

PROPOSITION 3 (TRIPLE CONSTRAINT ON STRONG ALGORITHMIC CAPTURE). *Complete strong algorithmic capture of consciousness requires the simultaneous satisfaction of:*

*(1) Representability:* $\mathrm{Truth}(C) \subseteq \mathcal{R}$*,*

*(2) Algorithmic realizability of explanatory relations,*

*(3) Phenomenal explanatory closure.*

*Failure of any one of these conditions is sufficient to block the existence of a theory* $T \in \mathcal{A}$ *such that* $\mathrm{Capture}_{\mathrm{str}}(T, C)$ *with full coverage of* $\mathrm{Truth}(C)$.

JUSTIFICATION. These conditions correspond respectively to: (i) the formal availability of the explanandum, (ii) the computational realizability of explanatory relations, and (iii) the satisfaction of the explanatory demand characteristic of the hard problem.

Each condition corresponds to a distinct requirement for complete explanation:

- Without representability, the explanandum cannot be formally specified.
- Without algorithmic realizability, the explanatory relations cannot be implemented within $\mathcal{A}$.
- Without explanatory closure, the resulting theory fails to explain why experience exists, even if it describes its structure.

Since strong capture requires all three conditions, the failure of any one suffices to block it. □

## 11.5 Interpretation and Temporal Scope

These three layers jointly support $A2'$ without requiring the stronger claim that $\mathrm{Truth}(C) \notin \mathcal{A}$.

The resulting position is conditional but robust, since failure at any one of the three constraints is sufficient to block complete strong capture:

- If phenomenal truths are not fully representable, complete capture fails.
- If representation is possible but not algorithmically realizable, complete capture fails.
- If both hold but explanatory closure fails, complete capture still fails.

It is important to be explicit about the temporal scope of these claims. The stronger thesis $A2$ — that $\mathrm{Truth}(C) \notin \mathcal{A}$ — is a claim in principle. If correct, it holds regardless of future theoretical developments: no algorithmic theory, present or future, could achieve complete strong capture.

The weaker operational premise $A2'$ is not a claim of that kind. It is supported by the current state of the field: no existing algorithmic theory has been shown to satisfy all three necessary conditions simultaneously. Future developments could, in principle, undermine $A2'$ by establishing representability, algorithmic realizability, and explanatory closure together. The triple constraint identifies the specific conditions that would need to be met.

Thus, the distinction between $A2$ and $A2'$ corresponds to a distinction between a principled impossibility and a current-state limitation:

- $A2$ asserts that the limitation is structural and cannot be overcome.
- $A2'$ asserts that the limitation has not been overcome, and that the conditions required to overcome it are individually unestablished and jointly demanding.

The main theorem holds under $A2'$. If a future theory satisfies all three constraints, it would refute $A2'$ and the theorem would no longer apply. But it would do so by achieving precisely what the present paper identifies as missing: a complete algorithmic account that combines representability, realizability, and phenomenal explanatory closure. The paper therefore does not foreclose this possibility. It identifies what it would require.

## 11.6 Evidential Direction: From A2' Toward A2

There is, however, an evidential asymmetry between the two premises that should be made explicit.

If $A2$ is true, then $A2'$ follows immediately: a principled impossibility entails a current-state failure. Conversely, if $A2$ is false — if complete strong algorithmic capture is in principle achievable — then the persistent failure to satisfy any of the three necessary conditions, despite sustained effort across multiple disciplines, calls for an independent explanation.

The triple constraint identifies three conditions that must hold simultaneously. Each has been individually challenged from different and largely independent directions:

- **Representability** is challenged from the philosophy of mind: if phenomenal truths are essentially perspectival or first-personal, they may resist the kind of third-person formal encoding that algorithmic theories require [40, 44].
- **Algorithmic realizability** is challenged from computability theory and cognitive science: the Church–Turing thesis is not formally provable [15], the algorithmic status of human cognition is underdetermined (Proposition 1), and the inference from local to global computability fails (Layer 3).
- **Explanatory closure** is challenged from the phenomenology of the explanatory gap: no existing theory — including GWT, IIT, and recurrent processing theory — has been

shown to explain why its identified mechanisms are accompanied by experience rather than occurring without phenomenal character (Section 9.5).

These three lines of failure are *independent*. The representability challenge arises from the structure of phenomenal facts. The realizability challenge arises from formal constraints on computation. The explanatory closure challenge arises from the nature of the explanatory demand itself. They do not reduce to a single underlying difficulty, nor does progress on one automatically yield progress on the others.

This convergence is evidentially significant. The hypothesis that $A2$ is true provides a unified explanation for why all three conditions fail simultaneously: they fail because $\text{Truth}(C)$ is not the kind of object that admits complete algorithmic capture. The hypothesis that $A2$ is false — that complete strong capture is achievable but has simply not yet been achieved — must treat each failure as a separate contingent gap, with no common explanation for their co-occurrence.

On standard principles of inference to the best explanation, a hypothesis that unifies independently observed phenomena provides stronger evidential support than one that treats each as coincidental. Accordingly, $A2'$ is not evidentially neutral between $A2$ and $\neg A2$. The pattern of evidence that supports $A2'$ — systematic, multi-dimensional, and independently motivated failure — is more naturally explained by the truth of $A2$ than by the hypothesis that all three conditions will eventually be satisfied but have simply not been met yet.

A natural response to this argument is that the failure may reflect the youth of consciousness science rather than a principled limitation. Modern empirical consciousness research dates only to the early 1990s, and significant progress requires time.

This concern is worth taking seriously. However, the pattern of progress within the field is itself evidentially relevant. A recent comprehensive review describes the current state of consciousness science as an "uneasy stasis," noting over two hundred distinct theoretical approaches exhibiting wide divergence in metaphysical assumptions and explanatory strategies, with most empirical research designed to support rather than test or compare theories [14]. The most rigorous adversarial test to date — the COGITATE collaboration — found that its results challenged key tenets of both leading theories rather than favoring either [16]. Kuhn's taxonomy identifies approximately 225 theories organized into ten categories spanning quantum mechanics to cosmic consciousness [34].

This is not the profile of a field that is merely young. It is the profile of a field that is empirically productive but explanatorily fragmented. If the problem were primarily one of insufficient data, one would expect theoretical convergence as data accumulates. Instead, theoretical divergence has increased alongside empirical progress. This asymmetry — advancing weak and structural capture alongside stalled explanatory closure — is precisely what the present framework predicts and is more naturally explained by a structural barrier at the level of strong capture than by a temporary data deficit.

The argument developed here does not depend on a claim that further progress is impossible. It depends on the observation that progress has followed a specific and persistent pattern: success at lower levels of capture without convergence at the level of explanatory closure. This pattern may eventually be broken, but its persistence across three decades of intensifying empirical work constitutes evidence that should not be dismissed as mere immaturity.

To be clear: this is an inductive observation, not a deductive proof. It is possible that future work will satisfy all three conditions and thereby refute both $A2'$ and $A2$. The present paper does not claim otherwise. But it is worth stating explicitly that the evidential weight of $A2'$ does not distribute equally between the two possibilities. The structure of the failure points toward a principled limitation rather than a temporary one.

The main theorem requires only $A2'$. But the broader philosophical significance of the result is shaped by how one reads the relationship between the two premises: as a temporary gap awaiting closure, or as the visible surface of a deeper structural limitation.

### 11.7 What Would Be Required to Refute A2'

To reject $A2'$, a reductionist account must demonstrate all of the following:

(1) That $\text{Truth}(C)$ is fully representable within a formal system,
(2) That the mapping between phenomenal and physical or functional descriptions is algorithmically derivable,
(3) That this mapping yields genuine explanatory closure rather than redescription,
(4) That structural reproduction suffices for phenomenal instantiation.

At present, none of these conditions has been conclusively established.

### 11.8 Summary

The main theorem expresses a conditional boundary on explanation grounded in the current structure of the problem, rather than a purely stipulative constraint.

The failure of complete strong algorithmic capture is not attributable to a single limitation, but to the convergence of representational, computational, and explanatory constraints. These constraints are independently motivated and do not reduce to one another. While the theorem requires only the weaker premise $A2'$, the pattern of independent failures that supports $A2'$ is itself evidentially asymmetric: it is more naturally explained by the principled thesis $A2$ than by the hypothesis that each failure is a separate contingent gap. This does not establish $A2$, but it indicates the direction in which the available evidence points.

## 12 Formal Framework and Assumptions

Let:

- $\mathcal{A}$: algorithmic theories
- $\mathcal{E}$: admissible explanations
- $C$: phenomenal consciousness (as defined above)
- $\text{Truth}(C)$: non-explanatory truths about $C$

We are concerned specifically with theories that aspire to complete strong capture of consciousness.

Assumptions:

$$A1 : \mathcal{E} \subseteq \mathcal{A}$$
$$A2' : \neg\exists T \in \mathcal{A} \text{ such that } \mathrm{Capture}_{\mathrm{str}}(T, C)$$
$$A3 : \text{A theory completely explains } C \text{ only if } \mathrm{Capture}_{\mathrm{str}}(T, C)$$

## 13 Main Theorem

The theorem below is not intended as an independent discovery derived from minimal premises. Its deductive structure is straightforward. The substantive burden of the paper lies in motivating $A2'$ and clarifying why complete explanation requires strong capture in the sense defined above.

Theorem 4. *If A1, A2', and A3 hold, then no algorithmic theory completely explains phenomenal consciousness in the sense of complete strong capture.*

The plausibility of assumption $A2'$ is supported by the Triple Constraint on Strong Algorithmic Capture established above. Given $A2'$, the main result follows straightforwardly. The significance of the theorem lies in the justification of its premises rather than in the complexity of the deduction.

Proof. Assume, for contradiction, that there exists some theory $T \in \mathcal{E}$ that completely explains phenomenal consciousness.

By $A3$, complete explanation requires $\mathrm{Capture}_{\mathrm{str}}(T, C)$.

By $A1$, since $T \in \mathcal{E}$, it follows that $T \in \mathcal{A}$.

But by the definition of $\mathrm{Capture}_{\mathrm{str}}(T, C)$, strong capture requires that $\mathrm{Truth}(C) \subseteq \mathrm{Scope}(T)$.

Hence there exists an algorithmic theory $T \in \mathcal{A}$ such that $\mathrm{Capture}_{\mathrm{str}}(T, C)$ and $\mathrm{Truth}(C) \subseteq \mathrm{Scope}(T)$, contradicting $A2'$.

Therefore, no such theory exists. □

## 14 Synthesis, Simulation, and Instantiation

The question of whether consciousness can be built is distinct from the question of whether it can be explained. Even if complete strong explanation fails, it may still be possible to predict, simulate, reproduce, or instantiate consciousness. These possibilities must be carefully distinguished.

### 14.1 Four Levels of Artificial Realization

Four increasingly strong notions should be distinguished.

*Prediction.* A system predicts the behavior, reports, or outward signatures associated with consciousness without thereby reproducing the relevant internal organization.

$$\mathrm{Pred}(S, C)$$

Prediction is the weakest notion. It is compatible with mere behavioral mimicry and therefore provides no sufficient evidence of consciousness [4, 56, 57].

*Simulation.* A system simulates consciousness when it models the dynamics or organization of a conscious system in a representational or emulative way.

$$\mathrm{Sim}(S, C)$$

Simulation is stronger than prediction, but does not by itself imply that the simulated phenomenon is instantiated. A simulation of digestion does not digest; likewise, a simulation of consciousness need not be conscious [25, 58, 59].

*Structural reproduction.* A system structurally reproduces consciousness when it realizes the relevant organization of a conscious system rather than merely representing it.

$$\mathrm{Struct}(S, C)$$

This notion is stronger than simulation. It concerns the implementation of the relevant architecture, dynamics, or causal organization.

*Instantiation.* A system instantiates consciousness when phenomenal consciousness is genuinely present in that system.

$$\mathrm{Inst}(S, C)$$

Instantiation is the strongest notion and is the one ultimately at stake in debates about artificial consciousness.

### 14.2 Logical Relations

These notions form a hierarchy:

$$\mathrm{Inst}(S, C) \Rightarrow \mathrm{Struct}(S, C) \Rightarrow \mathrm{Sim}(S, C) \Rightarrow \mathrm{Pred}(S, C)$$

but not conversely.

In particular:

$$\mathrm{Pred}(S, C) \not\Rightarrow \mathrm{Inst}(S, C)$$

and

$$\mathrm{Sim}(S, C) \not\Rightarrow \mathrm{Inst}(S, C)$$

These non-implications are central to the present paper.

### 14.3 Prediction and the Mimicry Problem

There is broad agreement that behavioral prediction is insufficient for consciousness. A system may produce outputs that are indistinguishable from those of a conscious subject while lacking any experience at all. This concern appears in classical form in Searle's Chinese Room argument [57], and has been sharpened in recent work on artificial intelligence.

Bengio and Elmoznino [4] warn explicitly against the risk of mistaking behavioral competence for genuine consciousness, noting that systems trained on human-generated data will naturally reproduce the outward signatures of conscious agents. Schwitzgebel [56] develops a related "mimicry argument," according to which sufficiently advanced AI systems will inevitably produce outputs that appear conscious regardless of whether they possess any inner experience.

Similarly, Birch [5] frames this as a "gaming problem": systems may pass behavioral or functional tests for consciousness while implementing fundamentally different internal processes.

Accordingly, the following implication fails:

$$\mathrm{Pred}(S, C) \not\Rightarrow \mathrm{Inst}(S, C)$$

Even perfect behavioral success does not by itself establish the presence of phenomenal consciousness.

## 14.4 Simulation and Structural Reproduction

The deeper debate begins once one moves beyond prediction to systems that reproduce more than input-output behavior.

A computational-functionalist position holds that if the right functional organization is realized, then consciousness is present. On this view, sufficiently faithful structural reproduction yields instantiation. Chalmers' organizational invariance arguments are the canonical defense of this position [12]. In abstract form, the claim is:

$$\mathrm{Struct}(S, C) \Rightarrow \mathrm{Inst}(S, C)$$

A biological-naturalist position denies this implication. On this view, reproducing functional organization is insufficient because the causal powers relevant to consciousness are biological or substrate-dependent. Searle's critique of computationalism [58] provides the classical formulation, arguing that formal or computational structure alone cannot generate genuine mental phenomena. More recently, Seth [59] defends a biologically grounded account on which consciousness depends on the properties of living systems, suggesting that artificial systems would need to differ substantially from current computational architectures in order to be conscious.

On this picture, simulation and even structural reproduction can fail to instantiate consciousness, in much the same way that a simulation of digestion does not itself digest.

Thus, the implication

$$\mathrm{Struct}(S, C) \Rightarrow \mathrm{Inst}(S, C)$$

is precisely one of the central disputed claims in the literature.

## 14.5 Substrate, Architecture, and Causal Organization

The literature suggests that objections to artificial consciousness operate at multiple levels. Some objections concern algorithms, others concern architecture, and still others concern the physical substrate itself. This multi-level structure is explicitly developed in recent taxonomic work on digital consciousness [41, 42], which distinguishes objections arising at the level of computation, causal organization, and physical realization.

This motivates a finer decomposition of structural reproduction:

$$\mathrm{Struct}(S, C) \in \{\mathrm{Struct}_{\mathrm{func}}, \mathrm{Struct}_{\mathrm{causal}}, \mathrm{Struct}_{\mathrm{sub}}\}$$

where the relevant notion of structure may be functional, causal, or substrate-bound.

The functionalist position tends to privilege:

$$\mathrm{Struct}_{\mathrm{func}}(S, C)$$

while biological and IIT-style objections often require stronger conditions involving causal or substrate-level equivalence.

## 14.6 Integrated Information and Non-Equivalence of Functional Simulation

Some theories explicitly reject the inference from functional equivalence to conscious equivalence. In particular, Integrated Information Theory (IIT) holds that consciousness depends on the intrinsic causal structure of a system rather than merely on its input-output behavior [1, 62]. On this view, two systems may be functionally equivalent while differing in their degree of integrated information ($\Phi$), and therefore in their level of consciousness. In particular, if consciousness depends on integrated informational or causal structure rather than merely on input-output behavior, then two functionally equivalent systems may differ in consciousness. This is one way of understanding the claim that a simulation of a conscious architecture need not itself be conscious, even if it perfectly reproduces behavior.

Formally, this means that there may exist systems $S_1, S_2$ such that:

$$\mathrm{Pred}(S_1, C) = \mathrm{Pred}(S_2, C)$$

while:

$$\mathrm{Inst}(S_1, C) \neq \mathrm{Inst}(S_2, C)$$

This possibility is one of the strongest reasons not to collapse simulation into instantiation.

## 14.7 Empirical Challenges to Computational Sufficiency

Recent empirical work has raised further challenges for purely computational accounts of consciousness. Larkum et al.[37] investigate the relationship between neural activity and computational structure, identifying cases in which ongoing neural dynamics appear to diverge from the underlying computational organization.

Their results suggest a possible disconnect between computation and phenomenology, raising the possibility that even structurally faithful computational models may fail to capture the processes relevant to conscious experience. While these findings do not decisively refute computational theories, they reinforce the need to distinguish carefully between simulation, structural reproduction, and instantiation.

## 14.8 Non-Computable and Non-Classical Positions

A stronger anti-functionalist position holds that consciousness depends on non-computable or non-classical physical processes. Penrose and Hameroff represent the most prominent version of this view [49, 50]. If such views are correct, then no classical computational system can instantiate consciousness, regardless of how accurate its simulation may be.

This yields the stronger claim:

$$C \notin \mathcal{A} \Rightarrow \forall S \in \mathcal{A},\ \neg\mathrm{Inst}(S, C)$$

The present paper does not assume this stronger thesis, but it remains one live possibility within the broader debate.

### 14.9 Agnosticism and the Epistemic Gap

A principled agnostic position holds that we currently lack any decisive method for determining whether artificial structural reproduction yields phenomenal instantiation. Birch [5] argues that this uncertainty is unavoidable given the "gaming problem": any behavioral or functional test for consciousness may be satisfied by systems that lack genuine experience.

Related concerns appear in recent interdisciplinary work on AI consciousness, which emphasizes the need for theory-guided indicators rather than purely behavioral criteria [9, 10]. These approaches aim to identify structural or functional markers associated with consciousness, but they do not resolve the deeper question of whether such markers are sufficient for phenomenal instantiation.

This leads to the epistemic limitation:

$$\neg\mathrm{Provable}\big(\mathrm{Struct}(S,C) \Rightarrow \mathrm{Inst}(S,C)\big) \land \neg\mathrm{Provable}\big(\mathrm{Struct}(S,C) \not\Rightarrow \mathrm{Inst}(S,C)\big)$$

Thus, even in the presence of increasingly sophisticated artificial systems, the question of whether consciousness is instantiated may remain underdetermined.

### 14.10 Relation to the Present Framework

The distinctions introduced here clarify the scope of the paper's main theorem.

The theorem does not rule out:

- predictive success,
- empirical modeling,
- weak capture,
- or even some forms of structural reproduction.

What it targets, conditionally, is complete strong algorithmic explanation.

Thus, even if artificial systems can simulate, reproduce, or perhaps instantiate consciousness, it would not follow that phenomenal consciousness has been completely explained in the strong sense defined earlier. The following section develops a related point at the level of human judgment: observers may attribute consciousness to systems that exhibit only predictive or conversational success, thereby collapsing distinctions that the present framework is intended to preserve.

### 14.11 A Structural Limitation Result

The fragmentation considerations discussed earlier also matter here. Even if each component of consciousness is individually computable, the whole need not be algorithmically capturable as a unified object:

$$(\forall i,\ c_i \in \mathcal{A}) \not\Rightarrow C \in \mathcal{A}$$

This means that successful simulation of parts of consciousness does not by itself imply complete synthesis of the whole.

The transition from local computability to global instantiation remains one of the deepest unresolved issues in the theory of consciousness.

### 14.12 Indicator-Based Approaches

Recent work proposes shifting the focus from behavioral tests to theoretically grounded indicators of consciousness. Butlin et al. [9] develop a framework in which candidate systems are evaluated against properties derived from leading neuroscientific theories, such as global workspace dynamics, recurrent processing, and integrated information.

These approaches aim to bridge the gap between prediction and structural reproduction by identifying features that are plausibly necessary for consciousness. However, they do not eliminate the distinction between structural reproduction and instantiation, and therefore remain compatible with both functionalist and biological interpretations of consciousness.

## 15 Attribution of Consciousness to Artificial Systems

The preceding section established a hierarchy of realization levels and corresponding non-implications, in particular:

$$\mathrm{Pred}(S,C) \not\Rightarrow \mathrm{Inst}(S,C)$$

This result is purely formal. However, the possibility it identifies is not merely theoretical. There is now substantial empirical evidence that human observers systematically treat prediction-level success as evidence of phenomenal consciousness.

### 15.1 Empirical Pattern: Attribution from Prediction-Level Behavior

Recent large-scale studies show that the conflation between behavioral success and consciousness attribution is widespread. Colombatto and Fleming [19] report that in a nationally representative sample, roughly one-third of participants denied that ChatGPT has subjective experience, while approximately two-thirds attributed at least some degree of phenomenal consciousness. These attributions are graded rather than binary, but a majority of respondents crossed the minimal threshold for attributing phenomenal experience.

Crucially, these attributions are not driven by structural or mechanistic understanding. Rather, they scale with usage frequency and are predicted by perceived experiential qualities rather than by perceived intelligence or reasoning ability. Complementary findings are reported by Kang et al. [32], who show that metacognitive self-reference and the expression of apparent emotions significantly increase perceived consciousness, whereas emphasis on knowledge or factual competence reduces it.

Taken together, these results indicate that what triggers consciousness attribution is not evidence of structural reproduction or instantiation, but surface-level behavioral features characteristic of conversational fluency and self-presentation. In the terms of the present framework, these are features associated with:

$$\mathrm{Pred}(S,C)$$

rather than with stronger forms of realization.

## 15.2 Mechanism: Anthropomorphism and Attribution

The observed pattern admits a well-supported psychological explanation. Humans exhibit a general tendency to attribute mental states to non-human systems under conditions of sufficient behavioral complexity and social interaction. This tendency has been extensively documented in the literature on anthropomorphism and social cognition [26, 45].

Contemporary AI systems amplify this effect. Pataranutaporn et al. [47] describe large language models as *anthropomorphic conversational agents* that are capable of producing language indistinguishable from human communication, thereby triggering what they term *anthropomorphic seduction.* Seth [59] further analyzes this phenomenon as the result of two interacting biases: anthropocentrism, which treats human consciousness as the reference model, and anthropomorphism, which projects human-like characteristics onto non-human systems.

Within this framework, the attribution of consciousness is best understood as a property of the observer's cognitive processes rather than as evidence of any underlying phenomenal state in the system. This can be expressed schematically as:

$$\mathrm{Attribution}(O, S, C) \not\Rightarrow \mathrm{Inst}(S, C)$$

where $O$ denotes an observer. This is a stronger claim than the formal non-implication between prediction and instantiation. It shows that the inference from behavior to consciousness fails not only at the level of logical entailment, but also at the level of the psychological mechanisms that generate the inference itself.

## 15.3 Consequences for Explanation

These findings bear directly on the explanatory project developed in this paper.

First, the prevalence of consciousness attribution to systems exhibiting only prediction-level behavior shows that behavioral evidence cannot serve as a reliable indicator of strong capture. If observers attribute phenomenal consciousness on the basis of conversational fluency, then any criterion for explaining consciousness that relies on behavioral or folk-psychological indicators will be systematically too permissive. In particular, success at:

$$\mathrm{Pred}(S, C)$$

may be misinterpreted as evidence for:

$$\mathrm{Inst}(S, C) \quad \text{or even} \quad \mathrm{Capture}_{\mathrm{str}}(T, C)$$

despite the absence of explanatory closure.

Second, there is a more general epistemic risk. As artificial systems become increasingly effective at simulating the outward signatures of conscious agents, the distinction between prediction, structural reproduction, and instantiation may become progressively harder to detect in practice. Under such conditions, the question of whether consciousness has been explained risks becoming entangled with the question of whether it has been attributed. This does not alter the formal structure of the argument developed in this paper, but it does affect the empirical context in which that argument is evaluated.

In particular, the conditional failure of complete strong explanation may become difficult to recognize, as systems that satisfy all observable behavioral criteria give rise to the appearance of explanatory success without satisfying the conditions for strong capture.

The framework developed here is partly motivated by precisely this risk. By defining complete explanation in terms of strong capture rather than behavioral adequacy, it separates the question of explanation from the psychology of attribution, ensuring that the standard for explaining consciousness cannot be met by predictive success alone, however compelling it may appear to observers.

# 16 Discussion

The preceding analysis identifies multiple independent sources of limitation. These limitations are not only theoretical. As the previous section shows, they also structure the practical problem of how human observers interpret artificial systems, often conflating predictive adequacy with phenomenal presence:

- Computational limitation: the possibility that complete explanation exceeds algorithmic derivation,
- Representational limitation: the possibility that phenomenal truths are not fully formalizable,
- Conceptual limitation: the explanatory gap between structure and experience,
- Cognitive limitation: the underdetermined algorithmic status of human reasoning,
- Explanatory limitation: the gap between weak or structural success and strong capture,
- Instantiational limitation: the gap between simulation, structural reproduction, and phenomenal presence,
- Attributional limitation: the tendency of observers to infer consciousness from behavior in ways that outrun the evidential basis for instantiation,
- Scope limitation: the paper's results are conditional on phenomenal realism; under illusionism, the target of explanation dissolves rather than being shown to resist explanation.

Taken together, these constraints suggest that the obstacle to complete explanation is not a single isolated barrier. The problem instead appears at the convergence of several distinct limitations, each of which blocks a different route to complete strong capture. The attributional limitation is especially important in the context of artificial intelligence, because it can make the failure of strong explanation difficult to detect in practice: systems that satisfy behavioral expectations may generate the appearance of consciousness, and even the appearance of explanatory success, without meeting the conditions required for instantiation or strong capture.

## 16.1 Objection: Functional Sufficiency

A natural objection is that functional organization is sufficient for consciousness, and therefore that sufficiently detailed computational models could in principle achieve complete strong capture of $C$. On such a view, the weaker operational premise A2' is false, and the main theorem does not apply. More strongly, such a position would also undermine the broader philosophical thesis A2 [12, 52, 54].

The present paper does not refute this position outright. Instead, it clarifies what must be established in order for such a position to succeed. A defender of functional sufficiency must show not merely that functional structure tracks consciousness, but that it yields phenomenal explanatory closure and full coverage of the relevant truths.

## 16.2 Objection: Type-B Physicalism and the Explanatory Gap

A more sophisticated reductionist position is given by type-B physicalism. On this view, the explanatory gap between physical descriptions and phenomenal consciousness is accepted as genuine, but is interpreted as a limitation of human cognition rather than as evidence of a metaphysical distinction. Consciousness is held to be identical to physical processes, even if this identity is not transparently explainable.

Levine's formulation of the explanatory gap emphasizes precisely this point: there is a gap between physical or functional accounts and phenomenal experience that is not merely epistemic but explanatory [38]. Type-B physicalists accept this gap while denying that it implies any ontological separation.

This position poses a direct challenge to the present framework. If the explanatory gap is cognitive rather than metaphysical, then the failure to achieve explanatory closure may reflect limitations of human understanding rather than limitations of algorithmic theories.

The present framework is compatible with this interpretation. The limitation result does not depend on whether the explanatory gap is metaphysical or cognitive in origin. If explanatory closure is required for complete explanation (A3), and if such closure is systematically unavailable to cognitive agents, then complete explanation remains unavailable in practice, regardless of the source of the limitation.

Thus, even under type-B physicalism, the conditions for complete strong capture are not satisfied. The limitation therefore applies at the level of explanation as such, rather than at the level of metaphysical interpretation.

## 16.3 Objection: The Moving Target Problem

A further concern is that the target of explanation may itself be unstable. As empirical and theoretical work progresses, the characterization of consciousness may change, potentially altering what counts as a complete explanation [13, 64].

On this view, the apparent failure of complete explanation may reflect not a structural limitation, but an evolving target. If $\mathrm{Truth}(C)$ is not fixed, the requirement of full coverage may seem indeterminate or perpetually revisable.

The present framework does not require that all truths about consciousness be known in advance. It requires only that a complete explanation, if one exists, must in principle range over the full truth-set of phenomenal consciousness, however that set is ultimately characterized. The limitation result therefore concerns any sufficiently rich and stable notion of $\mathrm{Truth}(C)$ and does not depend on premature closure of the empirical or conceptual domain.

## 16.4 Objection: Scope and Modesty

Another objection is that the notion of complete explanation adopted here is excessively strong. One might argue that scientific explanations are typically partial, domain-specific, and revisable, and that requiring full coverage of $\mathrm{Truth}(C)$ sets an unrealistically high bar [60, 65].

This objection is correct if the claim under consideration were merely that science should produce useful or progressively better accounts of consciousness. But that is not the claim at issue here. The paper distinguishes weak, structural, and strong forms of capture precisely in order to preserve the legitimacy of partial and empirically grounded explanations.

The limitation result is intentionally narrower. It targets a stronger ambition: that phenomenal consciousness can be completely explained in a way that closes the explanatory gap rather than merely correlating with, modeling, or functionally organizing its contents. The distinction is particularly important in the case of artificial systems, where behavioral and social fluency can easily encourage over-ascription of consciousness without resolving the underlying explanatory problem.

## 16.5 Objection: The Meta-Problem Dissolves the Hard Problem

Chalmers [13] introduces the *meta-problem of consciousness*: the problem of explaining why we think there is a hard problem. Even if the hard problem itself resists solution, the meta-problem may be tractable. A theory might explain why cognitive systems generate reports about an explanatory gap, produce intuitions of phenomenal irreducibility, and find consciousness puzzling, all without needing to explain phenomenal consciousness itself.

This raises a potential objection to the present paper. If the meta-problem can be solved within an algorithmic framework, then the appearance of an explanatory gap would be fully accounted for, and the residual demand for strong capture might be dismissed as a cognitive artifact rather than a genuine limitation.

The present framework has a natural response. The meta-problem concerns why we *judge* that there is a hard problem. In the terms adopted here, it targets truths about cognitive and functional processing:

$$\mathrm{Meta}(C) \subseteq \mathrm{Func}(C)$$

A solution to the meta-problem would explain why systems like us produce reports and intuitions about consciousness. But it would not thereby explain why phenomenal consciousness exists. In formal terms:

$$\mathrm{ExplClosure}(T, \mathrm{Meta}(C)) \nRightarrow \mathrm{ExplClosure}(T, C)$$

Explaining why we *think* there is an explanatory gap is not the same as closing the gap. A theory that accounts for our problem-intuitions achieves a form of functional or structural capture of the meta-problem, but this does not constitute strong capture of phenomenal consciousness itself.

Thus, the meta-problem is compatible with, and does not undermine, the central limitation developed here. It shows that certain aspects of the *discourse* about consciousness are algorithmically tractable, while leaving the target of the present paper — the explanation of phenomenal consciousness — untouched.

### 16.6 Objection: Illusionism and the Status of $C$

A more radical objection holds that phenomenal consciousness, as defined in this paper, does not exist. Illusionism, most prominently defended by Frankish [28], maintains that our sense of having phenomenal experiences is due to systematic introspective misrepresentation: experiences seem to have phenomenal properties, but this seeming is itself a functional state, not a phenomenal one. On this view, the hard problem dissolves because there is nothing for it to be about.

If illusionism is correct, then $C$ as defined in this paper — phenomenal consciousness not exhausted by functional accessibility — does not exist, and $\mathrm{Truth}(C)$ is either empty or ill-defined. The entire framework developed here would lose its target.

The present paper does not attempt to refute illusionism. Instead, it treats phenomenal realism — the view that there is something it is like to be in certain states, irreducible to functional organization — as a scope condition. The paper's results hold conditional on phenomenal realism, which remains the majority position in philosophy of mind [6, 12].

Two observations are worth noting, however. First, illusionism does not eliminate the explanatory burden; it relocates it. The "illusion problem" — explaining why experiences systematically seem to have phenomenal properties they lack — faces its own explanatory difficulty. As Kammerer [31] argues, accommodating a systematic misrepresentation of neural activity to an introspecting subject appears no easier than accommodating phenomenal consciousness itself. The second-order problem may be at least as hard as the first-order one.

Second, the meta-problem analysis developed in the preceding subsection applies directly to illusionism. A theory that explains why we *judge* ourselves to be phenomenally conscious does not thereby establish that phenomenal consciousness is illusory. It establishes only that the judgment is functionally explicable. Whether the judgment is also *accurate* — whether there really is phenomenal consciousness — remains a separate question that the meta-problem alone does not settle.

Accordingly, illusionism represents a position under which the paper's target dissolves, rather than a position that refutes the paper's claims within their intended scope.

### 16.7 Relation to Existing Formal Theories of Consciousness

The formal vocabulary introduced in this paper — levels of capture, realization hierarchy, representability, and strong capture — operates at a different level from the formalisms employed by object-level theories of consciousness.

Theories such as Integrated Information Theory [1, 62], Global Workspace Theory [2, 23], and higher-order theories [55] each propose specific mechanisms, quantities, or architectures as the basis of consciousness. IIT defines a measure of integrated information ($\Phi$); GWT characterizes a broadcasting architecture; higher-order theories posit meta-representational states. Each of these is an *object-level* theory: it makes claims about what consciousness is or how it arises.

The present framework does not compete with these theories. It is *meta-theoretic*: it provides a vocabulary for evaluating what any such theory achieves in terms of explanatory completeness. The question addressed here is not which object-level theory is correct, but whether any theory of a certain formal type (algorithmic) can satisfy the conditions for complete strong capture.

This distinction matters because the limitation result developed in this paper is compatible with the success of any particular object-level theory at the level of weak or structural capture. A theory may be empirically well-supported, predictively powerful, and structurally illuminating while still failing to achieve explanatory closure. The meta-theoretic framework makes this possibility precise without prejudging which object-level theory best describes the mechanisms of consciousness.

Accordingly, the present paper should not be read as an argument against any specific neuroscientific or computational theory. It is an argument about the *type* of explanatory achievement that algorithmic theories can, conditionally, be shown to lack.

### 16.8 Limitations

The present analysis has several important limitations.

First, the main result is conditional on assumptions A1–A3, in particular the operational premise A2'. The paper does not establish that complete strong algorithmic capture is impossible in principle, but rather analyzes the consequences of its current absence and the conditions required for its realization.

Second, the framework does not refute reductionist positions such as functionalism or physicalism. Instead, it clarifies what must be established for such positions to achieve complete explanatory success in the strong sense defined here.

Third, the argument depends on phenomenal realism as a scope condition. If phenomenal consciousness, as defined, does not exist, then the target of explanation dissolves rather than being shown to resist explanation.

Finally, the paper does not resolve questions concerning the instantiation of consciousness in artificial systems. It distinguishes levels of realization and attribution, but remains agnostic on whether artificial systems can genuinely instantiate phenomenal consciousness.

## 17 Conclusion

Explaining consciousness may fail at multiple levels: representational, structural, computational, cognitive, instantiational, and attributional. In particular, predictive or structural success need not amount to phenomenal instantiation, and phenomenal instantiation need not amount to complete strong explanation. Contemporary artificial systems make this distinction practically urgent, because they already elicit robust attributions of consciousness on the basis of behavioral fluency alone.

The result established here is conditional: if no algorithmic theory can achieve complete strong capture of phenomenal consciousness, then complete algorithmic explanation is impossible. The weaker premise $A2'$ on which this result depends is supported by the current state of the field, specifically by the independent failure of representability, algorithmic realizability, and explanatory closure. The paper identifies the specific conditions whose joint satisfaction would be required to overcome this limitation.

While the theorem does not require the stronger principled thesis $A2$, the evidential structure of the argument is not neutral between the two readings. The convergence of three independently motivated failures is more naturally explained by a structural limitation than by three coincidental gaps. This observation does not constitute a proof of $A2$, but it indicates the direction in which the available evidence points. The algorithmic status of human reasoning itself remains unresolved, reinforcing the conditional nature of any claim about the limits of explanation.

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